

Gavin Cheung

gavin.cheung12@gmail.com | (978)-505-3187 | US Citizen | [linkedin.com/in/gavcheung](https://www.linkedin.com/in/gavcheung) | [personal website](#)

EDUCATION

Rensselaer Polytechnic Institute

Troy, NY

B.S in Computer Science (focus in AI & ML, Minor in Cognitive Science)

Expected May 2028

Coursework: Software Principles, Machine Learning, Operating Systems, Data Structures and Algorithms

EXPERIENCE

Network, Security, and Data Analytics Intern - Enterprise & Tactical Networking Team

June 2026 - Present

MITRE

Bedford, MA

- Shipped **full-stack** features within a **~1M LOC** production codebase, reducing AWS cloud costs by **~33%** for **1,000+** concurrent remote sessions across **30+ government agencies**.
- Pitched final product to project **sponsors**, division leadership, and full-time engineers. Presented with a company award in recognition of outstanding contribution to the project and mission.
- Placed **1st** and 3rd out of two company-wide **hackathons** (AI & Autonomous Robotics), collaborating in **interdisciplinary teams** to build working prototypes under tight deadlines.

Machine Learning and Back End Developer

Aug 2025 - January 2026

Rensselaer Center for Open Source

Troy, NY

- Developed data pipeline to process **40,000+** in-game scenarios through a series of Scikit-Learn random forest models.
- Engineered **2,400+ lines** of code across 21 commits to an **open-source ML** sports data project, focusing on feature engineering and UI/UX redesign to achieve 90%+ model accuracy and smooth user experience.
- Managed project lifecycle using Scrum/Agile methodologies, version control, documentation, and software principles.

Research Technology Intern

May 2025 - Aug 2025

HARP Research Inc.

Troy, NY

- Cut inference benchmarking time for Reduced Order Models to under 3 seconds across **10,000+ nodes** by performance-profiling Python code to **identify latency bottlenecks**.
- Reduced system configuration time by **~55%** by containerizing a multiphysics simulation environment with Docker
- Co-authored a technical paper on optimization methods by synthesizing research on Dynamic Assembly algorithms and quantifying performance **trade-offs** in Topology Optimization

IT Assistant

May 2024 - Aug 2024

Veterans Association Medical Center

Bedford, MA

- Designed and implemented an automated barcode scanning system, streamlining the expired medication return process and **increasing operational speed by over 300%**.
- Identified and **capitalized \$10,000+** in expired drug credit through database matching and expiration tracking.

PROJECTS

Phishy URL Detection Ensemble - 3rd place @ MITRE AI Hackathon | Python, ScikitLearn, Pandas, Kaggle, GitLab, Optuna

- Achieved 98.6% accuracy and 0.041 log loss classifying 1,000+ URLs by stacking 9 machine learning models (XGBoost, CatBoost, LightGBM, SVM, Naive Bayes) into a regularized meta-learner
- Prevented overfitting and reduced training time by integrating AI tools (Claude Code, Codex) into feature engineering workflow, validating with repeated k-fold cross-validation, and automating hyperparameter search with Optuna

LEADERSHIP & INVOLVEMENTS

SEO Tech Developer LITE Program | SEO | Remote

- Selected for competitive technical program, completing coursework across core software engineering fundamentals: Bash/Linux, REST APIs, Git version control, databases, OOP, unit testing & CI/CD pipelines, software design, etc.

Python Instructor | coding&&community | Troy, NY

- Deliver weekly Python lessons to 7th–12th graders, simplifying complex concepts into an accessible curriculum.

TECHNICAL SKILLS

Languages/Libraries: Python, C/C++, PostgreSQL, Java, JavaScript, HTML/CSS, Pandas, ScikitLearn, Flask

Tools: Git, VSCode, React, Docker, GitLab, Github, DBeaver, Codex, Claude Code,